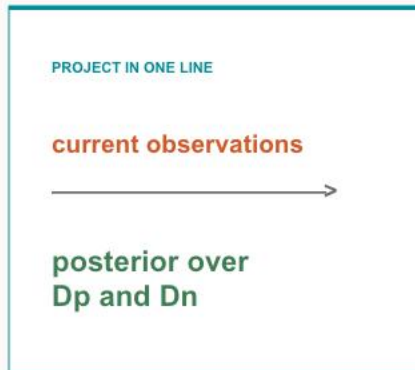


Current-Based Bayesian Inference for a 1D Poisson-Nernst-Planck System

A parametric physics-informed neural network as an amortized forward surrogate

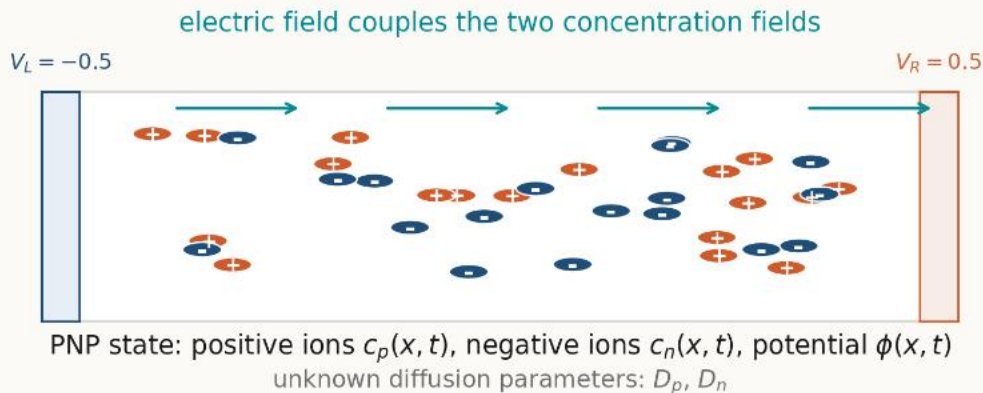
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PNP is a coupled model of charged particles and an electric field.

The inference target is physical: the diffusion rates of the positive and negative species.



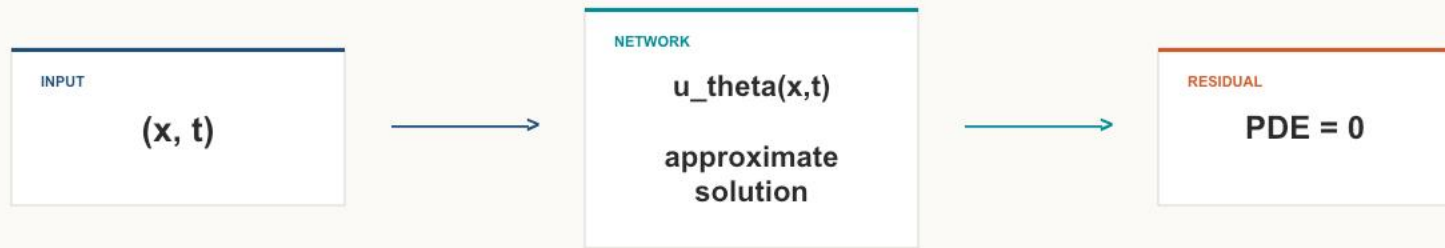
The bottleneck is repeated forward solves over parameter space.

A Bayesian inverse problem may need many candidate evaluations, not just one accurate simulation.



A PINN trains the solution function by penalizing physics violations.

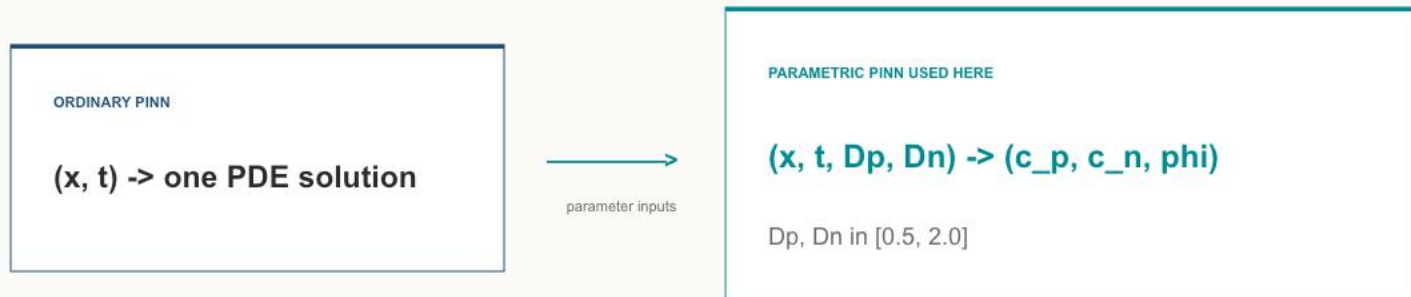
The network is not given a full table of solution labels; the governing equations enter the loss.



$$\mathcal{L}(\theta) = \lambda_{PDE} \mathcal{L}_{PDE} + \lambda_{IC/BC} \mathcal{L}_{IC/BC} + \lambda_{coverage} \mathcal{L}_{param}$$

Adding D_p and D_n to the input turns one network into a family of solvers.

This is the step that makes the trained PINN useful for inverse inference.



- Candidate parameters become ordinary neural-network inputs.
- The inverse problem can evaluate many candidates without retraining.

The benchmark is controlled: 1D, nondimensional, and blocking-electrode.

$$J_p = -D_p \partial_x c_p - D_p Z + c_p \partial_x \phi, \quad J_n = -D_n \partial_x c_n - D_n Z - c_n \partial_x \phi$$

$$\partial_t c_p + \partial_x J_p = 0, \quad \partial_t c_n + \partial_x J_n = 0$$

$$-\varepsilon \partial_{xx} \phi - (Z + c_p + Z - c_n) = 0$$

The equations couple diffusion, electric drift, conservation of each species, and Poisson electrostatics.

ACTIVE SETUP

x in $[0, 1]$, t in $[0, 0.2]$

D_p, D_n in $[0.5, 2.0]$

$\phi(0,t) = -0.5$, $\phi(1,t) = 0.5$

$J_p = J_n = 0$ at both walls

Blocking electrodes: ions do not cross the boundary, but charging current can still be measured.

Training builds the surrogate; the reference solver is not used as a label set.

The reference model is held for validation and synthetic current generation.

PHYSICS

PDE residuals

CONSTRAINTS

**initial and boundary
conditions**

COVERAGE

sampling over D_p , D_n

$$\mathcal{L}(\theta) = \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}} + \lambda_{\text{IC/BC}} \mathcal{L}_{\text{IC/BC}} + \lambda_{\text{coverage}} \mathcal{L}_{\text{param}}$$

500k

training steps

The inverse problem is posed around current, not hidden concentration fields.

The PINN predicts fields internally, but the likelihood compares a measurable signal.



79 current intervals, 2% relative Gaussian noise, uniform prior on the parameter box.

The measured quantity is $I(t)$; $Q(t)$ is a stable simulator-side construction.

In the blocking-electrode setup, current is charging current through the external circuit.

EXPERIMENTAL VIEW

**external circuit measures
current $I(t)$**

$$Q_L(t) = -\varepsilon \frac{V_R - V_L}{L} - \int_{x_L}^{x_R} \frac{x}{L} \rho(x, t) dx$$

$$I_k = \frac{Q_L(t_{k+1}) - Q_L(t_k)}{t_{k+1} - t_k}$$

Using $Q(t)$ avoids differentiating the learned potential exactly at the wall.

This is a modeling choice for the synthetic benchmark, not an assumption that $Q(t)$ is directly measured.

Gaussian current errors turn residuals into a posterior over diffusion parameters.

The current diagnostic uses a transparent grid posterior before moving to heavier samplers.

$$I_{\text{obs},k} = I_{\text{PINN},k}(D_p, D_n) + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, \sigma^2)$$

$$p(D_p, D_n \mid I_{\text{obs}}) \propto \exp \left[-\frac{1}{2} \sum_k \left(\frac{I_{\text{PINN},k}(D_p, D_n) - I_{\text{obs},k}}{\sigma} \right)^2 \right] p(D_p, D_n)$$

41 x 41

parameter grid

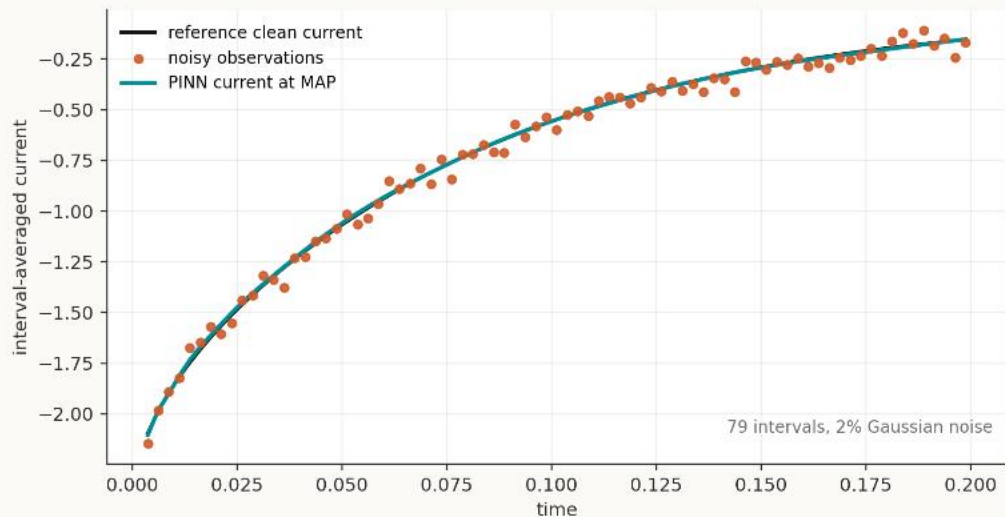
uniform

prior over box

2%

relative current noise

At the MAP point, the PINN-predicted current follows the noisy observations.

**1.25**

MAP Dp

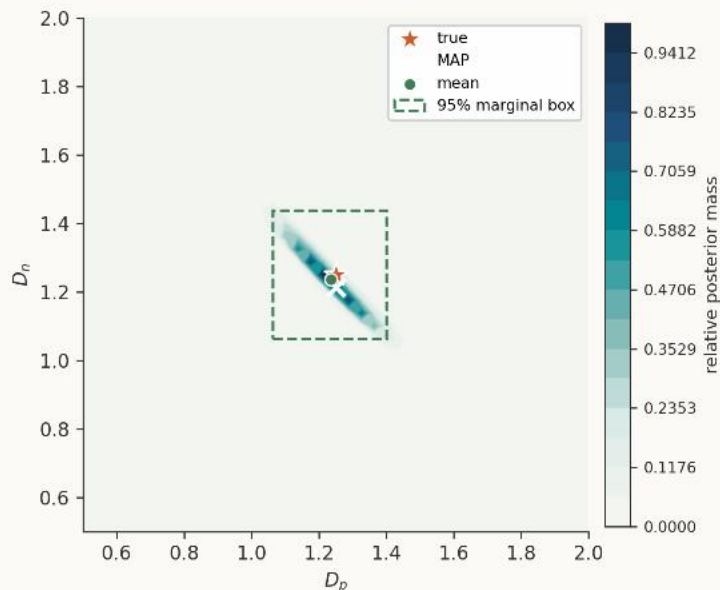
1.2125

MAP Dn

79

current intervals

The posterior covers the truth and reveals coupled uncertainty between D_p and D_n .



SUMMARY

true: $D_p = 1.25$, $D_n = 1.25$

posterior mean: $D_p = 1.2353$, $D_n = 1.2368$

95% marginal intervals

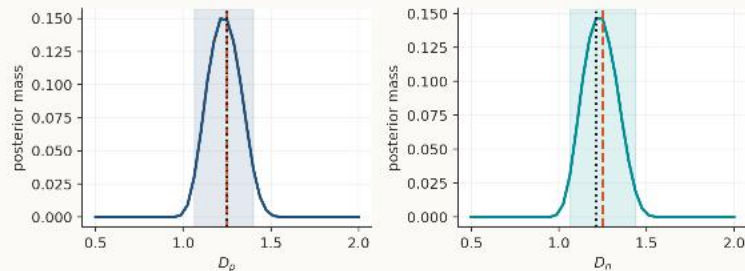
D_p : [1.0625, 1.4000]

D_n : [1.0625, 1.4375]

Truth lies inside both intervals.

The posterior shape is part of the scientific result.

The problem is indirect, coupled, and sensitive to both observation noise and surrogate error.



- The data are current only; c_p , c_n , and ϕ are not directly observed.
- D_p and D_n affect the signal through a coupled nonlinear PDE.
- The forward map is approximate, so validation against the benchmark remains necessary.

A narrow-looking curve fit is not enough; the posterior shows what the current data actually identify.

A current-based PINN inverse pipeline is working in a controlled benchmark.

WHAT WORKS

Current-only synthetic data recover a posterior region that contains the true D_p and D_n .

WHAT THE PINN DOES

It amortizes the forward PNP map across parameter space; it is not the inverse estimator by itself.

WHAT COMES NEXT

Stress-test more true parameters, noise levels, sampling schedules, and then replace the grid by MCMC.

Main message: train the forward surrogate once, then use measurable current to quantify uncertainty in diffusion parameters.

Any questions?

Current-based Bayesian inference for a 1D PNP system using a parametric PINN surrogate